

Question Number	Answer	Additional Guidance	Mark
2ai	An answer that includes four of the following points: • use of {water culture / soilless compost} (1) • with and without VCL (1) • control of appropriate biotic variable (1) • control of appropriate abiotic variable (1) • remove roots (1) • dry to constant mass (1)	accept vermiculite / perlite e.g. same {variety / type / cultivar} of tomato plant starting condition {same size / age of seedling / plant} seed source ignore species e.g. temperature / pH / light intensity / suitable time for growth (if a time is quoted must be at least a day) / concentration of Hoagland ignore soil accept known in any case accept amount for time only accept a description	4

Question Number	Answer	Additional Guidance	Mark
2aii	A calculation that includes the following steps. • {correct bars identified / difference found} (1) • correct division performed correctly and answer multiplied by 100%	e.g. {115 AND 95 / 115 - 95 / 20} $20 \div 95 = 0.21$, $0.21 \times 100 = 21\%$ accept 21.05 and 21.1 accept 21.0 for 1 mark	2

Question Number	Answer	Additional Guidance	Mark
2bi	An answer that includes the following points: Length • hard to judge where {top / bottom} of shoot is / shoot {width may vary / may be branched / may be bent} (1) Fresh shoot • (fresh mass will include) water content which will lead to variability of mass of tissue (1)	accept reverse argument	2

Question Number	Answer	Additional Guidance	Mark
2bii	A calculation showing the following steps. • correct measurement of {XY / length of shoot} from photograph (1) • correct conversion of measurement from mp1 to actual length (1)	inspect photograph for any sign of measurement e.g., 38 (mm) accept any value between 37 and 38 (mm) accept if given in cm if correct $38 \times (50 \div 5) = 380$ (mm) Accept value between 370 and 380 correct answer with no working gets 2 marks	2

Question Number	Answer	Additional Guidance	Mark
2biii	An answer that includes five of the following point: • VCL enhances {growth / dry mass / shoot length} when both K and P absent • VCL has very small effect on {growth / dry mass / shoot length} when N is absent (1) • VCL shows biggest (percentage) increase when no P (1) • VCL improves root growth (in all treatments) (1) • comment on lack of any data to allow comment on significance of any differences (1) • reference to {cost / sustainable / renewable} (1)	accept VCL enhances {growth / dry mass / shoot length} accept comparative words , longer, heavier etc. accept no effect accept VCL contains K and P but not N for mps 2 and 3 do not accept reliability do not accept climate change	5